II. Give very short answer to the following questions.

17. Define student's t-statistic.
18. Define: degrees of freedom.
19. Define the paired t-statistic.
20. When paired t-test can be applied?
21. Write the test statistic to test the difference between normal population means.
22. Write the standard error of the difference between sample means.
23. Define chi-square statistic.
24. Write the applications of chi-square distribution.
25. What are the minimum requirements of chi-square test?
26. Define an attribute.
27. Give the recurrence formula for binomial distribution.

III. Give short answer to the following questions.

28. List out the properties of t-distribution.
29. Write down the applications of t-distribution.
30. Explain the testing procedure to test the normal population mean, when population variance is unknown.
31. Write down the procedure to test significance for equality of means of two normal populations based on small samples.
32. A random sample of ten students is taken and their marks in a particular subject are recorded. The average mark is 60 with standard deviation 6.5. Test the hypothesis that the average mark of students is 55.
33. State the properties of χ^2 distribution.
34. What is a contigency table?
35. Write the procedure to test the population variance.
36. Write the test procedure for testing the independence of attributes.
37. Write down the computational steps for testing the significance of goodness of fit.
38. Give the test statistic for 2×2 contingency tables.

IV. Give detailed answer to the following questions.

39. A random sample of 10 packets containing cashew nuts weigh (in grams) 70,120,110,101, 88,83,95,98,107,100 each. Test whether the population mean weight of 100 grams?
40. The average run of cricket player from the past records is 80. The recent scores of the player in 6 test matches are 84, 82, 83, 79, 83 and 85. Test whether the average run is more than 80?
41. The heights (in feet) of 6 rain trees in a town A are 30, 28, 29, 32, 31, 36 and that of 8 rain trees in another town B are 35, 36, 37, 30, 32, 29, 35, 30. Is there any significant difference in mean heights of rain trees?

12th_Statistics_EM_Unit_2.indd 72 1/28/2020 7:10:17 PM